UA LITTLE ROCK INSTITUTIONAL REVIEW BOARD

For office use only
Protocol #:
Date Received:

The approval period for all protocols is one year. Eight weeks before the end of the approval period submit a continuation form (https://ualr.edu/irb/home/irb-forms/) to request approval to continue the research. We have updated other templates and forms. To avoid delay, use the most current form.

Request for Protocol Review

This form is to be used for ALL new protocols.

The UA Little Rock Institutional Review Board (IRB) must review all projects that involve human participants. The IRB will determine whether the project meets federal guidelines and constitutes human participants research. **The project may not start until IRB approval is granted**. If the duration of the project exceeds the length of the approval period, submit a Request for Continuation two months before approval expires.

Note: UA Little Rock policy requires that all persons listed on this document show successful completion of training in the protection of human participants in research. Please attach the Group 1 CITI training completion report for all listed persons. This protocol will not be reviewed until all CITI training records are attached. *Further instructions are in red.*

Principal Investigator: Kartik Sakthivel	Email: ksakthivel@ualr.edu
Department: Information Science	Phone: 603-493-9567
CITI Expiration date(s): **Jan 03 2025**	
Co-Principal Investigator:	Email:
Department:	Phone:
CITI Expiration date(s):	
Co-Principal Investigator:	Email:
Department:	Phone:
CITI Expiration date(s):	
Co-Principal Investigator:	Email:
Department:	Phone:
CITI Expiration date(s):	
Faculty Advisor/Chair of thesis or dissertation committee *(must be included if PI or Co-PI is a student)*: Dr. Elizabeth Pierce	
Department: Information Science	Email: expierce@ualr.edu
CITI Expiration date(s): **09/04/2024**	Phone: 501-916-5223
Project Title: An Industry Study Of Best Practices That Mitigate Potential Bias In Use Of External Data Sources In Life Insurance Automated Underwriting Models	
Is this project externally/internally funded?	No
If so, please indicate the funding source:	
Is this project requested by a third party? If so, please elaborate under item 1 below*	No
If sponsored, will data be owned solely by the sponsoring third party?	No
Is the data for this study already collected and available to the public? If data is derived from a proprietary data set, secondary analysis or any other combination, please elaborate under Item 2 below.	No
Description of Human Participants:	

Are participants members of a protected class? • Under 18 years of age • Confined in a correctional or detention facility	 No No
Will personal records/data be collected without informed consent?	No
If collected, will personal records/data be directly or indirectly identifiable?	No
Is any of the research conducted at a location other than UA Little Rock?	Yes
If yes, where: Conducted via electronic survey and subsequent telephonic/WebEx (or Zoom) video conference based interviews within the United States of America	

Provide the information requested below.

1. Describe the purpose of the project

*Provide a brief 1-2 paragraph description and explanation of the purpose of this project, including the significance of the research, and the research question(s). *(If a third party requested the project, also answer the following: Who is the requesting party? What is their request? What is the PI's relationship with third party?)*

This research will provide an industry-level view into the use of external data sources by insurance carriers that are established/mature in the automated/accelerated underwriting process, and the steps, if any, being undertaken to intercept and flag data elements that might introduce bias into the automated underwriting process. This research aims to provide a list of most commonly used external data sources, and which potential data elements within these data sources can inject bias into the automated underwriting process. No such research currently exists across the life insurance industry, and could help provide guardrails and parameters to smaller to mid-size firms, and the industry as a whole, on how to intercept and mitigate potential sources of bias and proxy discrimination.

This industry-level research will pursue the following objectives:

Objective 1. Conduct an up-to-date current state assessment of external data sources for automated underwriting by helping to identify the top external sources of data feeding automated underwriting models across the industry.

Objective 2. Perform an analysis of potential gaps in the automated and accelerated underwriting process. This gap analysis will be performed in two stages. The first step will be to provide this list of top external data sources to the industry, and flagging data elements from these common external data sources in automated underwriting that firms are on the lookout for due to concerns that these data elements could potentially introduce bias. This flagging of data elements will be based on an aggregation of what each carriers are currently doing, in addition to being informed by what other adjacent sectors are doing in dealing with the same data elements. The second step would be in identifying companies that are further along in their automated and accelerated underwriting maturity, with the goal being to understand how these firms are managing these data elements.

Objective 3. Provide insights into what actions, if any, firms considered mature in the process are currently undertaking to proactively identify data elements that could potentially introduce bias or proxy discrimination, why firms selected those specific data elements, how they are flagging these specific data elements, and what results they are seeing from doing so. This Objective will yield the development of industry-level best practices that can be adopted and adapted by all firms across the industry during their own automated and accelerated underwriting journey. The outcome of publishing these best practices would be that companies can learn, in a standard manner, what to focus on when using external data sources, while maintaining their own proprietary AI/ML underwriting models. In addition to the practical applicability of this Objective, this research will also inform additional, more specialized study and potentially lead to the development of an industry utility that is developed by an industry trade association such as LIMRA LOMA LL Global.

2. Describe the methods and procedures

Describe the data collection procedures and what participants will have to do: If potentially sensitive information is collected in an interview or if there is concern about participants' or the investigators' safety, please address procedures to protect the participants' confidentiality and/or participants' and investigators' safety during data collection. What happens to the data if a participant withdraws or skips questions?

IRB approval is being sought since the case study research will involve a survey that is fielded to willing participants across the Life Insurance industry (to fulfil Objectives 1 and 2, and seek volunteers for Objective 3) and shall then delve into interviews with human subjects after survey results are compiled (to fulfil Objective 3).

LIMRA & LOMA is the largest and oldest non-advocacy trade association for Life, Group Life, Group & Worksite Benefits, Annuities, Financial Services and Retirement in the world. LIMRA, based in Windsor, CT, is a 103-year-old not-for-profit organization primarily focused on industry leading benchmarking and research. LIMRA provides industry accredited training programs for the aforementioned sectors of insurance. LOMA, based in Atlanta, GA, is a ~ 100-year-old for-profit organization focused on the same industry verticals. LOMA provides industry accredited, coveted and prestigious designations for the insurance industry. 99.8% of North American insurance companies are members of the association. Both LIMRA and LOMA, having merged as one association 13 years ago, and bringing together 200+ years of industry knowledge, convene several industry conferences and events globally each year as well as host committees and study groups. LIMRA and LOMA operate in 71 countries, with over 1200 members.

1. Industry Voluntary Survey

LIMRA LOMA LL Global operates 124 committees and study groups in North America. One of the 124 study groups is one focused on Data and Analytics. Operating under the Center of Excellence for Data and Analytics, in partnership with the University of Connecticut (UCONN) and MIT, this study group is comprised of approximately 75 senior technology and analytics leaders (Chief Data Officers/Chief Data and Analytics Officers, VPs and higher from various Data and Analytics divisions) from a majority of the US and Canadian Life Insurance, Group Benefits, and Retirement companies. Similarly, the association also convenes a group of Chief Information Officers and Chief Technology Officers (or commensurate job title), as well as a group of Chief Underwriting Officers.

A voluntary survey, developed using Qualtrics, will be fielded to these constituencies of individuals across the industry. This survey will be distributed to the LIMRA Data and Analytics Study Group, and the LIMRA IT CxO Forum (comprised of approximately 50 CIOs and CTOs) across insurance firms, and the Chief Underwriting Officers Committee (comprised of approximately 30 Chief Underwriting Officers).
This survey and expected results will help to fulfil Objectives 1 and 2 of this research, while identifying voluntary participants for interviews to help fulfil Objective 3. The majority of data collection for this research will be via creation, distribution, and collection of the survey, after IRB approval. In total, it is expected that the survey will be fielded to somewhere between 100 to 150 industry leaders across the three aforementioned groups.

Since the survey response data is collected anonymously and participation is purely voluntary, non-completion of the survey means that responses will just not be recorded due to non-submission of the completed survey. At the end of survey, the questionnaire asks individuals to self-select in participating in a deeper set of interview questions (to fulfil development of best practices contributing to Objective 3 of this research). If individuals choose to participate and subsequently decide to withdraw participation, they will simply not be contacted for further dialog. If they decide to withdraw part-way through the interview process, any data collected (in terms of notes) will be destroyed. Note that these notes are going to be on paper and the notes will be shredded and disposed securely.

2. Interviews

Part of the survey questionnaire asks individuals to self-select in participating in a deeper set of interview questions that are germane to their organization, and specific to how their organization is handling potential bias in automated underwriting. Interviews shall be conducted via video conferencing or in-person as dependent on in-person meeting possibility due to geographies and/or travel restrictions due to the ongoing COVID-19 pandemic in 2022.
Interviewees are free to involve a maximum of three other individuals within their organization, those they deem as subject matter experts. Should these interviewees involve their peers in the interview, names and contact information of these individuals will be requested and obtained in advance. The interviews will not be recorded, and implicitly, automated transcription of these interviews via video conferencing platforms such as Cisco WebEx will be disabled. The notes from these interviews, as well as answers to questions, shall be recorded on paper. These notes,

after cleansing and formatting, will be supplied back to the interviewee/s in an email to ensure their feedback has been appropriately captured. The email address used to correspond with these individuals will be the researcher's UALR.edu domain email address (Gmail). A schedule of interviews shall be developed, along with a comprehensive roster of who will be interviewed, when, and how (medium). This schedule and associated data shall be provided to the IRB since there are human interviewees involved. It is expected that there is at least a 10% interview volunteer group of the 100 to 150 individuals the survey is sent to. A good count of interviewees is between 5 and 10, representing best practices of firms with established and more mature automated underwriting practices.

How long will this take participants to complete?

1. Survey – Survey will be in the field of a period of 6 weeks, with completion request reminders to be sent at the end of the 4th and 5th weeks. The survey itself should take no more than 20 minutes to complete at most.
2. Interviews – The interviews, for those who self-select to be interviewed on the survey, should take no more than 60 minutes. Interviews, for those volunteers, will be scheduled within four weeks after receipt of survey results.

Will follow-ups or reminders be sent? If so, explain (address the frequency, the modality, and attach a copy of the message):

Survey completion request reminders will be sent at the end of the 4th and 5th weeks of the survey having been fielded.

3. Describe the recruiting procedures *(Please attach copies of all recruitment material such as fliers, ads, phone scripts, emails, posts, etc.)*

Who will be recruited? Who are the participants? If this is an adult sample, how are the PI(s) verifying that participants are 18 or older? If applicable, how are PI(s) informing participants they must be 18 or older?

The research will be introduced at the Q2 2022 meeting of the Data and Analytics Study Group, the May 2022 convening of the IT CxO Forum, and the May 2022 convening of the Chief Underwriters Group. These groups will be notified about the nature, intent, and expected outputs and outcomes of this research. They will be informed that they should anticipate a survey via Qualtrics being sent to them, reiterating that participation is purely voluntary. At this stage, individuals will have a choice to simply not respond to the survey upon receipt, or explicitly ask that they not be sent the electronic survey – this is extremely unlikely. In this case, they will be removed from the survey distribution list.

How will participants be approached about participating in the study?

The survey and the reasoning behind the survey will be introduced at the respective gatherings of these groups as described above. The individuals within these groups will be sent the survey via Qualtrics. For those selecting to be interviewed, they will be contacted via email individually.

How will the names and contact information for participants be obtained?

The names and contact information of these participants is obtained via distribution lists for these individuals as appears in participation roster / email distribution lists, and as reflected in the LIMRA LOMA CRM. For those selecting to be interviewed, they will be asked to provide their name and email address (last section in the survey).

4. Describe compensation	
Will compensation be provided to participants?	*No*
If YES, please describe amount and type of compensation including money, gift certificates, extra credit, etc. If the study involves compensation through lottery, identify the odds of winning. If the study involves extra credit provide an explanation of the alternative activity participants are provided, and how much influence the extra credit might have on a student's grade.	
When will compensation be given? How much participation is required to earn the incentive? What happens if the participant quits before completion or skips some questions? Compensation must be prorated.	
Will compensation be paid by UA Little Rock?	*Choose Yes or No*

What participant information is required? The consent form should cover the information that is required for payment and the length of time their information is retained by the payment system.

How will you avoid compensation having a coercive effect on participants (i.e., that they will feel compelled to participate to earn compensation)? How will the participants' identity be kept confidential while allowing for compensation or extra credit to be awarded?

5. Conflict of Interest

State any financial or other relationships that are held by you or any individuals or institutions involved in the research which could create perceived or actual conflict of interest:
Serving as global Chief Information Officer of LIMRA LOMA LL Global, I will be working alongside my peer CIOs, CTOs, and CDAOs across the industry. Although not a perceived or an actual conflict of interest, to ensure maintenance of the highest levels of unwavering integrity and ethics which we live by, it will be made explicitly clear that this research, although will be for the industry, is being pursued by me in my capacity as a researcher and PhD candidate, and not on my capacity of Chief Information Officer.

State whether you or any individuals or institutions involved in the research will receive any compensation other than a grant award: No compensation will be received for this study.

Describe what reasonable and appropriate actions you plan to take to protect participants from the influence of the above conflict of interest:
During the introduction of the research to the various groups, and intimation that a survey will be sent to them, it will be made explicitly clear that this research, although will be for the industry, is being pursued by me in my capacity as a researcher and PhD candidate, and not on my capacity of Chief Information Officer.

6. Benefits and Risks

Explain the benefits to participants or to others. State benefits in probabilistic terms such as may or might (i.e., may lead to better services). There must be some benefits either to the participants (direct benefits) or to others (indirect benefits).
This research shall provide the industry best practices on how to successfully and confidently leverage external data sources in the process of automated underwriting, learning from established firms more mature in their journey, to ensure mitigation of potential bias due to these external data sources. LIMRA research indicates that most carriers have invested heavily in ensuring that automated and accelerated underwriting models are free of bias, and have implemented robust internal quality control protocols (Kamath, Clark, Purushothaman, & Sakthivel, 2021), there is yet to be research that identifies how established Life Insurance carriers are working on preventing biases from creeping into their Automated Underwriting models from internal and external data sources. These insights could be quite significant across the industry for all carriers to learn best practices from each other, without revealing their proprietary underwriting algorithms. Carriers hope that these automated underwriting programs, intended to provide a seamless customer buying experience, will entice more people to gain financial security and protection with a life insurance policy. Carriers, especially early adopters, and those further along in their AI/ML maturity, also recognize that the discovery of suspected bias, introduced accidentally in underwriting models within one organization, can be detrimental to all carriers and the industry as a whole; with ramifications ranging from loss of goodwill, to governmental investigations, and regulatory and compliance pressures.
Larger and better funded firms within the industry, due to their access to financial capital and resources, have had the ability to invest in developing robust automated and accelerated underwriting practices. With their diligence to ensure that their automated underwriting practices are free of any potential sources of bias and proxy discrimination, these firms could be considered as leaders in this emerging practice. The ability to invest significant amounts of resources and financial capital might not necessarily be possible for small or midsize insurance companies. According to the American Council of Life Insurers (ACLI), there are 840 life insurance companies operating in the United States as of the end 2020 (Alexander, Hoppenstedt, Sultanbikov, & Wang, 2021). These 840 companies are a combination of 747 US domestic life insurance carriers (including stock, mutual, fraternal, and other) and 93 foreign owned life insurance carriers operating within the United States. There are a significant number of firms considered to be in small to midsize segment that comprise a significant portion of these 840 companies. Managing a sizeable legacy

platform footprint, these companies do not have access to the financial resources required to achieve the same level of automated underwriting model maturity that the larger, better funded firms do, while also having to maintain their legacy technology platforms. The sizeable jump in the number of firms with some level of automated underwriting programs from 50% of those surveyed in 2017 (Art & Sondergeld, Transforming Underwriting: Automated Underwriting Company Practices for Life Insurance in 2017, 2017) to 84% (Kamath, Clark, Purushothaman, & Sakthivel, 2021) in 2021 according to LIMRA research indicates that there not only is a considerable interest across the industry, but there is also a "bandwagon" effect at play. Automated underwriting, across the industry, has evolved in a very short span of time from being implemented by roughly half the industry to becoming table stakes in the underwriting process today. With the recent introduction of external data sources to bolster the process, it is expected that external data sources for the purposes of automated and accelerated underwriting will become commonplace across the industry in the very near future. An oversight due to the lack of knowledge of best practices in the process of leveraging external data sources for the purpose of automated underwriting from any one of the 840 firms could pose the risk of eroding confidence in the entire industry. There is a significant coverage gap in the number of Americans protected with life insurance as part of sound financial planning, and where this gap is larger in communities of color. Absent guidance, inadvertent bias or proxy discrimination that occurs within any one firm can be detrimental to the industry's reputation especially at a time when the industry is seeking to protect as many American families with financial instruments such as life insurance.
The entire industry and every firm within it may benefit from this best practices case study.

Are there any known or anticipated risks to participants greater than those experienced in everyday activities? List any foreseeable physical and non-physical risks of participating in the study. Non-physical risks may include social, psychological, or economic harm; risks of criminal or civil liability; or damage to financial standing, employability, or reputation.	*No*
If YES, explain the risks.	
What will be done to minimize the risks? For projects that make a statement that counseling is not provided remember that there may be community resources available. Provide some examples. The Little Rock Community Mental Health Center is often a good solution. N/A	

7. Informed Consent (Please attach copies of informed consent /assent forms, emails, and/or letters. If participants are under 18 you must attach consent forms for parents or guardians.) Please describe your consent process, even if you believe this project is not human participants research and you have no formal consent form.

How will informed consent / assent be obtained?
Survey participation consent will be obtained via a checkbox at the outset of the survey. Interview consent will be implicitly obtained via responses to survey question asking if they would like to participate in an interview. Individuals have at that time, an option to select out from being interviewed.

*Are you requesting a **waiver of a signed consent form**?* *Providing participants with informed consent information is always required. However, requesting a **waiver of a signed consent form** may be appropriate for situations such as:* • *Low-risk, online, survey research, in which the participant consents to participate by checking a box; or* • *Sensitive research, in which a signed consent form may put the participant at undue risk.* *Tell us how participants can get a copy of the Informed Consent Form (i.e., screen shot, attached to recruiting e-mail, other).*	No

8. Confidentiality (Anyone who will collect, see or have access to the raw data must attach their certification document [Group 1 CITI Completion Report] attached to this protocol.)

How will participant confidentiality be maintained? Address how the participants' identity and data will be protected during data collection, analysis and when the data is reported. All response data will be collected as aggregated data and kept confidential. No PII will be collected or obtained unless respondent chooses to participate in the interview phase of the project. For respondents who agree to an interview, furnishing their names and email addresses, this

data shall be recorded by the researcher in a notebook/journal. This sheet shall be subsequently securely shredded after the interview is complete to maintain anonymity. Email exchanges that recap the dialog between the researcher and participant during the interview round shall be securely exchanged using the UALR Google Mail service (Gmail). These emails shall also be purged once the interviewee has reviewed the meeting minutes and agreed to what was discussed.

How will confidentiality of records be maintained? Address how the participants' identity will be protected when the data is reported. If you are using direct quotes, audio, or video clips, etc. these are identifying information and measures to protect confidentiality must be included. A non-exhaustive list of identifying information is at the end of this form. Combining categories will often identify an individual. Aggregating data may lower this risk when cell sizes are 5 or greater.

All survey data will be aggregate data and non-identifying. In the case of individuals who volunteer to be interviewed, data shall be recorded by the researcher in a notebook/journal. This sheet shall be subsequently securely shredded after the interview is complete (after Dissertation Chair review). Email exchanges that recap the dialog between the researcher and participant during the interview round shall be securely exchanged using the UALR Google Mail service (Gmail). These emails shall also be purged once the interviewee has reviewed the meeting minutes and agreed to what was discussed. The final research document shall make no references that would allow for firm or individual identification. The best practices research shall be genericized.

Who will be collecting the data? In most cases these individuals will need CITI training.
The researcher will be the only person collecting the data.

Who will be able to see or have access to the raw data? List everyone who has access to the raw data. This can include those who collect data, if they can see it, and those who analyze it. Submit CITI verification for anyone with access to the data.
Only the researcher, dissertation advisor, and one member of the dissertation committee will have access to aggregated raw data.

9. Data Security

How will data be reported? List all the ways the data will or might be disseminated. Also list whether the data will be reported individually or in aggregate. If aggregated, what is the smallest cell size that will be reported? If the project was requested by another party is the data reported by that party? If so how? To whom will data be reported? Are there plans to present at a conference or publish? Will data only be reported to a third party?

The final best practices case study research will be shared with the broader industry, including the aforementioned Data Analytics, IT CxO, and Chief Underwriters groups. There will be NO identifiable information in the final research. This research will be shared with organizations across the industry such that they can take action on these insights, and tighten their models to specifically ensure that they take mitigation steps to prevent this bias from infiltrating their automated algorithms. It is expected that this research will be globally be made available to LIMRA and LOMA's 1400+ members across 70+ companies via LIMRA.com. It is also foreseeable that an industry association like LIMRA LOMA LL Global could mutualize the problem solving by development of an industry utility that could ingest external data sources commonly used across the industry, weed out potentially troublesome data elements, and provide a cleansed or flagged data set to all carriers. For instance, if all carriers are using EHRs, LIMRA could build a utility to ingest EHRs, flag potential suspect data elements, and supply a relatively cleansed data set to all carriers, highlighting these suspect elements. This would avoid having each and every carrier develop their own flagging and cleansing tools and processes.
It is likely that a synopsis of this research will be published to the industry's trusted magazine, LIMRA MarketFacts as an article. Winner of the 2021 APEX Award for Publication Excellence, LIMRA MarketFacts has been publishing content that speaks to critical issues for the industry, including technology, retirement, insurance, workplace benefits,

globalization, financial wellness, employee productivity, professional development and assessment, operations, management, and products and services for over 60 years.
A copy of this research will also be made available to the LIMRA InfoCenter. The LIMRA InfoCenter serves as the corporate library for all insurance companies. The InfoCenter is staffed with professional research librarians who search LIMRA's research, as well as all industry and non-industry publications to find the answers members have. The InfoCenter is billed as "the first call members make to access information. Located in the William J. Mortimer library at LIMRA's headquarters in Windsor, Connecticut, the InfoCenter can serve as a vital resource to help make these best practices available to members of LIMRA LOMA LL Global.
Operating over 15 major conferences globally on an annual basis, and approximately 10 global webinars in a calendar year, it is anticipated that the findings and recommended best practices of this research will be presented in multiple forums as part of LIMRA LOMA LL Global. In addition to sessions on the topic at LIMRA LOMA conferences, it is anticipated that the research findings will be presented at various LIMRA LOMA CxO Consortia meetings, including but not limited to: the IT CxO Forum, the Chief Underwriters Forum, the Executive Roundtable, etc.

Where will the records be stored during data collection and analysis?

Aggregated survey results will be housed in Qualtrics. Qualtrics requires secure single sign-on and protects data-in-transit, as well as data-at-rest. Transmission is in compliance with appropriate SSL certificates and the highest level of compliance with the NIST guidelines. Interview results will be on paper, which shall be securely shredded after review of the Dissertation Chair, with email conversation recaps being on Google Mail (Gmail). UALR Gmail is end-to-end encrypted and can only be accessed via a UALR SSO Google Account login. No raw data shall be published or shared with anyone outside of the Dissertation Chair and one member of the Dissertation Committee.

Records must be kept for three years beyond the life of the study. Where will the records/data be kept and who will have access? If the PI is a student, the advisor must have access to the data.

Final data collected will be downloaded to a storage device (USB) that will be stored in a locked cabinet of the Department Chair of Information Sciences.

What security measures are in place to protect the data during data collection, data analysis, and the 3-year storage?

Data collected via Qualtrics will be behind SSO and aggregated results shall be stored in password protected format on a UALR-sanctioned server via Google Drive that is accessible to the Department Chair.

10. Attachment List:	
X	*CITI Training Records for all personnel. Do not send the certificate but rather the completion report that shows the completed modules. (The UA Little Rock policy on training in the protection of human participants can be found at http://www.ualr.edu/irb.)*
X	*Consent Forms and/or Assent Forms (See templates and guidelines at https://ualr.edu/irb/home/irb-forms/)*
X	*Site Letter of Support or Permission (Students must include a signed letter from the appropriate functionary at each site indicating that they are aware of the plan to conduct a study and what the study entails. Others must include a draft of such a letter.)*
X	*Recruiting Materials- The following abbreviated IRB statement can be included. "This study has been reviewed and approved by UA Little Rock's Institutional Review Board (IRB)." Make sure to put the PI's contact information on the recruitment materials.*
X	*Questionnaire, survey, etc. For online surveys attach a pdf or screen shots showing exactly what the participant sees. Make sure to include how the participant indicates consent.*

11. General Instructions:

A. All submitted protocols must be signed and dated.
B. Student research must be reviewed, signed and dated by the advisor.
C. Proofread the protocol before submission. Spelling and grammar errors can make a difference in meaning.
D. Make sure all titles and labels are consistent.
E. Attach all support documentation to the protocol by combining with this document into a single pdf file.
F. Include the IRB Approval statement in both the Informed Consent and Recruitment areas.
"This study has been reviewed and approved by UA Little Rock's Institutional Review Board (IRB). The IRB has determined that this study meets the ethical obligations required by federal law and University policies. If you have questions or concerns regarding this study, please contact the Principal Investigator (Provide the name of one or more researchers who can be reached for assistance. If you are a student, provide your advisor's name and contact information too. The IRB recommends that researchers not include home addresses, personal emails or private cell phone numbers). If you have any questions regarding your rights as a research participant, please contact the Research Compliance Officer at 501-916-6207 or irb@ualr.edu."
G. Send completed electronic submission to irb@ualr.edu.
H. To maintain proper record keeping, all communication with the IRB or the IRB Administrator must be through irb@ualr.edu .
I. Questions may be addressed to the IRB Administrator at irb@ualr.edu or by telephone at 501-916-6207.

12. Required Signatures:

Principal Investigator (PI) Signature and Date:

X	By checking this box and typing or signing my name and date below, I certify that the information provided above is correct and that this research will be conducted in accordance with federal regulations and UA Little Rock IRB policies and procedures on research with human participants.

Signature of Principal Investigator: Kartik Sakthivel **Date: *3/1/2022***
(Re-sign and date for every revision)

Advisor/Committee Chair Signature and Date (If PI is a student):
Advisor: After reviewing, please send completed protocol package from your UA Little Rock e-mail account to irb@ualr.edu.

☐	By checking this box and typing or signing my name and date below, I certify that I have reviewed and approve this student submission. I agree to supervise the research conducted by the student and I am aware that I am the responsible party for this research. I certify that this research will be the conducted in accordance with federal regulations and UA Little Rock IRB policies and procedures on research with human participants.

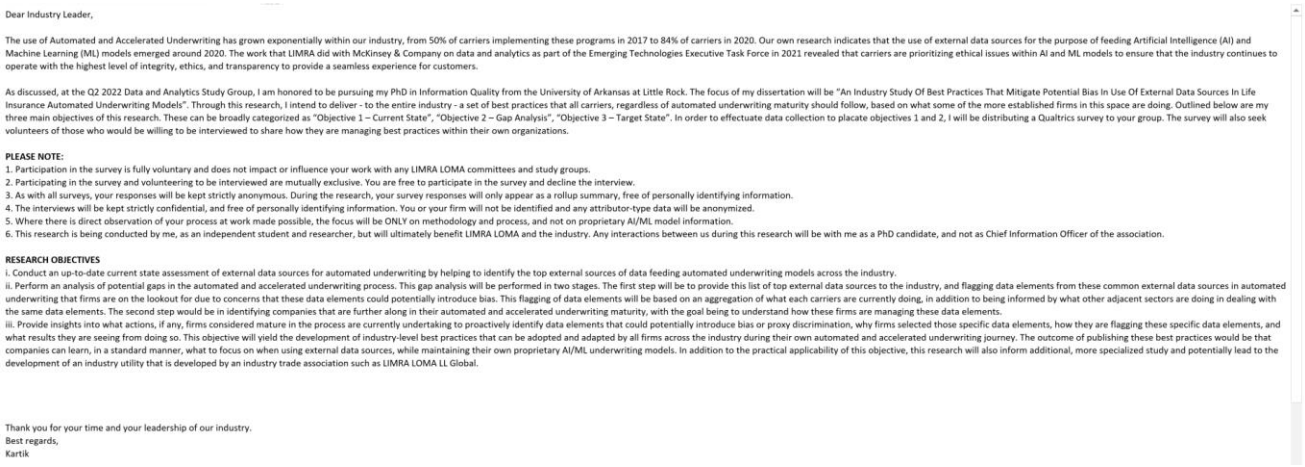
Signature of Faculty Advisor/Committee Chair: Elizabeth M. Pierce **Date:** 3/17/2022
(Re-sign and date for every revision)

Non-exhaustive List of Identifying information.

1. Names
2. All geographic subdivisions smaller than a state, including street address, city, county, precinct, zip code, and their equivalent geocodes, except for the initial three digits of a zip code if, according to the current publicly available data from the Bureau of the Census: (1) the geographic unit formed by combining all zip codes with the same three initial digits contains more than 20,000 people; and (2) the initial three digits of a zip code for all such geographic units containing 20,000 or fewer people is changed to 000.
3. All elements of dates (except year) for dates directly related to an individual, including birth date, admission date, discharge date, date of death; and all ages over 89 and all elements of dates (including year) indicative of such age, except that such ages and elements may be aggregated into a single category of age 90 or older.

4. Telephone numbers
5. Fax numbers
6. Electronic mail addresses
7. Social Security numbers
8. Medical record numbers
9. Health plan beneficiary numbers
10. Account numbers
11. Certificate/license numbers
12. Vehicle identifiers and serial numbers, including license plate numbers
13. Device identifiers and serial numbers
14. Web universal resource locators (URLs)
15. Internet protocol (IP) address numbers
16. Biometric identifiers, including finger and voice prints
17. Full face photographic images and any comparable images
18. Any other unique identifying number, characteristic or code

Email Notification to Potential Survey Participants



Dear Industry Leader,

The use of Automated and Accelerated Underwriting has grown exponentially within our industry, from 50% of carriers implementing these programs in 2017 to 84% of carriers in 2020. Our own research indicates that the use of external data sources for the purpose of feeding Artificial Intelligence (AI) and Machine Learning (ML) models emerged around 2020. The work that LIMRA did with McKinsey & Company on data and analytics as part of the Emerging Technologies Executive Task Force in 2021 revealed that carriers are prioritizing ethical issues within AI and ML models to ensure that the industry continues to operate with the highest level of integrity, ethics, and transparency to provide a seamless experience for customers.

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i. Conduct an up-to-date current state assessment of external data sources for automated underwriting by helping to identify the top external sources of data feeding automated underwriting models across the industry.
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Thank you for your time and your leadership of our industry.
Best regards,
Kartik

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EMAIL REMINDER 1 (DATA ANALYTICS GROUP, IT CxO Forum, Chief Underwriters)

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The survey can be access at: https://surveys.limra.com/jfe/form/SV_39u0ZeDZoOlNeN8

If you are interested in participating in this research and survey, gentle reminder that you have approximately two weeks before the survey closes.

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FINAL EMAIL REMINDER (DATA ANALYTICS GROUP, IT CxO Forum, Chief Underwriters)

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Survey (Text version)

Company Size: ( ) 100 – 500 | ( ) 501 – 1000 | ( ) 1001 – 5000 | ( ) 5001 – 10,000 | ( ) 10,0001+

1. On a scale of 1 to 5, 1 being least mature, and 5 being most mature, rank your firm's automated/accelerated underwriting models.

   1 | 2 | 3 | 4 | 5

2. Is your firm currently leveraging external data sources in your automated/accelerated underwriting process?

   Yes | No

3. If you answered No to 2, what are the timeframe expectations to incorporate external data sources in your automated/accelerated underwriting process?

   i. 6 months – 1 year

   ii. 1 year – 2 years

   iii. 2 years+

   iv. No plans

4. If you answered Yes to 1, rank the most common external data sources your firm is using, 5 being most used, and 1 being least used. Indicate 0 for data sources not on this list and name them.

   [ ] Credit History

   [ ] Criminal Records

   [ ] Medical Records / EHR

   [ ] MIB EHR

[ ] Milliman

[ ] Other PLEASE LIST

5. If you answered question 4 above, please indicate whether your firm has implemented procedures to flag data elements that could potentially be a source to introduce bias in the underwriting process.

   Yes | No | No – But it's in the works | No – We don't believe these data sources are a source of bias

6. If you answered Yes to question 5, please indicate level of maturity, 1 being least mature, and 5 being most mature.

   1 | 2 | 3 | 4 | 5

7. If you answered Yes to question 5, select Yes or No to submit to be considered for a detailed case study analysis. Analysis will include developing a grid of external data sources and potential data elements that might introduce bias into automated underwriting that your firm is currently flagging.

   Yes | No

8. If you answered Yes to question 7, please provide your name, organization, and email address below indicating your voluntary consent to be contacted for an interview.

   NAME:

   EMAIL ADDRESS:

   COMPANY:

Post-Survey 1-on-1 Interview Questions (Best Practices)

NAME/S AND TITLES (WILL BE ANONYMIZED):
COMPANY (WILL BE ANONYMIZED):

Q1. When did your firm first start implementing automated/accelerated underwriting?
Q2. What were the primary drivers behind your company pursuing automated/accelerated underwriting?
Q3. Assuming the initial implementation was germane to your firm's own data, what are some of the data considerations you had to keep at the forefront? Examples: staging pertinent data in one place, ensuring the data is of optimal quality for AI/ML models, data cleansing, data quality management, master and metadata management, data provenance, etc.
Q4. Were your AI/ML models developed in-house?
Q5. Knowing that firms are prioritizing transparency of models and ethical considerations before methodological and performance issues of models, how did your firm vet your own dataset?
Q6. Are you currently using external data sources for adjudicating underwriting during automated/accelerated underwriting?
Q7. What are the top external data sources that your firm is using? When did your firm decide to augment your own models and data with external data? Why? What was the rationale?
Q8. When considering ingesting of these data sources, what was the driving factor behind each and what went into the decision-making process to consume these specific data sources?
Q9. Within these data sources, what thought process went into ensuring these sources can be free of bias and proxy discrimination?
Q10. Within these data sources, are there specific data elements that your firm is paying additional attention to, in terms of evaluation or correction, to ensure your AI/ML models do not ingest data that could skew underwriting, or worse, introduce inadvertent bias or proxy discrimination into the model?
Q11. What kind of evaluation process exists within your firm when you considered these external data sources? How were potentially suspect data elements evaluated and discussed in this context?
Q12. What do you do when you suspect a data element within an external data source might introduce bias? Is there a scoring criteria for risk management on these elements?
Q13. Do you have a reciprocal process established to communicate back to the providers of this data if you do have questions regarding suspect data sources, or suspect with a degree of confidence that these data elements might actually be flawed and introduce bias?
Q14. Does your firm examine the use of the same data sets in other industries and evaluate what companies within those sectors are doing to mitigate potential bias being introduced into the system? Have you incorporated those lessons learned into your use of external data?
Q15. Do you have an internal governance board that reviews external data sources and deliberates usage of specific data source for underwriting? If so, who is on this governance board and how does the process work?

Q16. Do you have a risk scoring model developed for potential bias originating from existing data sources? How do you adjust your models to compensate if so?
Q17. What process exists in evaluating NEW data sources you plan to ingest into the model in the future?
Q18. If such a process exists, how does it work?
Q19. How do your models account for “Model Drift” in terms of having to be regularly tuned and maintained?
Q20. What do you foresee as the ultimate vision behind your company’s use of automated and accelerated underwriting?
QX: OPEN ENDED: Is there guidance or advice that you would like to share with the industry in your experience with managing external data sources for automated underwriting, or the whole process in general?

EXHIBIT D: Qualtrics Survey Link

https://surveys.limra.com/jfe/form/SV_39u0ZeDZoOlNeN8

Informed Consent

An Industry Study Of Best Practices That Mitigate Potential Bias In Use Of External Data Sources In Life Insurance Automated Underwriting Models

Introduction
You are invited to participate in a research project. In this research we are investigating/testing/evaluating best practices in the use of external data sources during the process of automated and accelerated underwriting across the life insurance industry. This study is being conducted by Kartik Sakthivel, as I am pursuing my PhD at the University of Arkansas at Little Rock (UA Little Rock). The Faculty advisor is Dr. Elizabeth Pierce, Chair of Information Science, UA Little Rock.

The purpose of this research is:
This research will provide an industry-level view into the use of external data sources by insurance carriers that are established/mature in the automated/accelerated underwriting process, and the steps, if any, being undertaken to intercept and flag data elements that might introduce bias into the automated underwriting process. This research aims to provide a list of most commonly used external data sources, and which potential data elements within these data sources can inject bias into the automated underwriting process. No such research currently exists across the life insurance industry, and could help provide guardrails and parameters to smaller to mid-size firms, and the industry as a whole, on how to intercept and mitigate potential sources of bias and proxy discrimination.

What is involved in the study?
Participation is completely voluntary. If you decide to participate you will be asked to complete a brief online survey that should take no more than 20 minutes to complete. The survey will also ask for volunteers who would choose to be interviewed as a part of the best practice research development. The interviews, for those who self-select to be interviewed on the survey, should take no more than 60 minutes. Interviews, for those volunteers, will be scheduled within four weeks after receipt of survey results.

Risks
This is a minimal risk research study. That means that the risks of participating are no more likely or serious than those you encounter in everyday activities.

The foreseeable risks or discomforts include the potential for loss of confidentiality, which is nearly always a risk in research studies. To minimize the risks for electronic records to be identifiable, the survey instrument will not collect your name, email, or identifying information. Should you submit for an interview, all conversations should be recorded in print, using a non-digital medium, and shall be destroyed after validation by the Department Chair.

Benefits
This research shall provide the industry best practices on how to successfully and confidently leverage external data sources in the process of automated underwriting, learning from established firms more mature in their journey, to ensure mitigation of potential bias due to these external data sources. LIMRA research indicates that most carriers have invested heavily in ensuring that automated and accelerated underwriting models are free of bias, and have implemented robust internal quality control protocols, there is yet to be research that identifies how established Life Insurance carriers are working on

preventing biases from creeping into their Automated Underwriting models from internal and external data sources. These insights could be quite significant across the industry for all carriers to learn best practices from each other, without revealing their proprietary underwriting algorithms. Carriers hope that these automated underwriting programs, intended to provide a seamless customer buying experience, will entice more people to gain financial security and protection with a life insurance policy. Carriers, especially early adopters, and those further along in their AI/ML maturity, also recognize that the discovery of suspected bias, introduced accidentally in underwriting models within one organization, can be detrimental to all carriers and the industry as a whole; with ramifications ranging from loss of goodwill, to governmental investigations, and regulatory and compliance pressures.

Larger and better funded firms within the industry, due to their access to financial capital and resources, have had the ability to invest in developing robust automated and accelerated underwriting practices. With their diligence to ensure that their automated underwriting practices are free of any potential sources of bias and proxy discrimination, these firms could be considered as leaders in this emerging practice. The ability to invest significant amounts of resources and financial capital might not necessarily be possible for small or midsize insurance companies that form a large portion of the 840 registered life insurance companies in the United States. These companies do not have access to the financial resources required to achieve the same level of automated underwriting model maturity that the larger, better funded firms do, while also having to maintain their legacy technology platforms. The sizeable jump in the number of firms with some level of automated underwriting programs from 50% of those surveyed by LIMRA in 2017 to 84% in 2021 according to LIMRA research indicates that there not only is a considerable interest across the industry, but there is also a "bandwagon" effect at play.

Automated underwriting, across the industry, has evolved in a very short span of time from being implemented by roughly half the industry to becoming table stakes in the underwriting process today. With the recent introduction of external data sources to bolster the process, it is expected that external data sources for the purposes of automated and accelerated underwriting will become commonplace across the industry in the very near future. An oversight due to the lack of knowledge of best practices in the process of leveraging external data sources for the purpose of automated underwriting from any one of the 840 firms could pose the risk of eroding confidence in the industry. There is a significant coverage gap in the number of Americans protected with life insurance as part of sound financial planning, and where this gap is larger in communities of color. Absent guidance, inadvertent bias or proxy discrimination that occurs within any one firm can be detrimental to the industry's reputation especially at a time when the industry is seeking to protect as many American families with financial instruments such as life insurance.

The entire industry and every firm within it may benefit from this best practices case study.

Confidentiality and Data Security

We will make every effort to keep information about you confidential. No PII will be collected or obtained unless respondent chooses to participate in the interview phase of the project. For respondents who agree to an interview, furnishing their names and email addresses, this data shall be recorded by the researcher in a notebook/journal. This sheet shall be subsequently securely shredded after the interview is complete to maintain anonymity. Email exchanges that recap the dialog between the researcher and participant during the interview round shall be securely exchanged using the UALR Google Mail service (Gmail). These emails shall also be purged once the interviewee has reviewed the meeting minutes and agreed to what was discussed. Please note that email communication is neither private nor secure. Though [I am/we are] taking precautions to protect your privacy, you should be aware that information sent through e-mail could be read by a third party.

Your confidentiality will be kept to the degree permitted by the technology being used. We cannot guarantee against interception of data sent via the internet by third parties. It is unlikely, but possible,

that others (UA Little Rock, or state or federal officials) may require us to share the information you give us from the study to ensure that the research was conducted safely and appropriately. We will only share your information if law or policy requires us to do so.

Audio/Video Recording
The interviews will not be recorded.

Future use of Identifiable Data or Specimens Collected in this Research

Compensation
There is no compensation for participating in this research.

Your rights as a participant
Please take whatever time you need to ask me questions about the study. You are free to stop participation at any time without any penalty. For the survey, if you do not wish to participate, please disregard the invitation for the survey. If you choose not to be interviewed, please indicate no on the survey. If you choose to be interviewed, but change your mind, please let me know via email.

IRB Approval and Contacts for questions or problems
This study has been reviewed and approved by UA Little Rock's Institutional Review Board (IRB). The IRB has determined that this study meets the ethical obligations required by federal law and University policies. If you have questions or concerns regarding this study, please contact the Principal Investigator (provide university email and university phone number if available) or Advisor (If there is an advisor provide university email and phone number if available). If you have any questions regarding your rights as a research participant, please contact the Office of Research Compliance at 501-916-6209 or irb@ualr.edu.

Consent of participant (or legally authorized representative)

If you are 18 years of age or older, understand the statements above, and freely consent to participate in the study, click on the "I Agree" button to begin the survey.

◯ I Agree ◯ I Do Not Agree

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)

COMPLETION REPORT - PART 1 OF 2
COURSEWORK REQUIREMENTS*

* NOTE: Scores on this Requirements Report reflect quiz completions at the time all requirements for the course were met. See list below for details. See separate Transcript Report for more recent quiz scores, including those on optional (supplemental) course elements.

- **Name:** Kartik Sakthivel (ID: 10645833)
- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
- **Institution Email:** ksakthivel@ualr.edu
- **Institution Unit:** Computer and Information Sciences - Information Quality
- **Phone:** 6034939567

- **Curriculum Group:** Human Research
- **Course Learner Group:** Group 1.Social and Behavioral Research Investigators and Key Personnel
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 45783192
- **Completion Date:** 04-Jan-2022
- **Expiration Date:** 03-Jan-2025
- **Minimum Passing:** 100
- **Reported Score*:** 100

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
Belmont Report and Its Principles (ID: 1127)	20-Dec-2021	3/3 (100%)
History and Ethical Principles - SBE (ID: 490)	20-Dec-2021	5/5 (100%)
Defining Research with Human Subjects - SBE (ID: 491)	03-Jan-2022	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	03-Jan-2022	5/5 (100%)
Assessing Risk - SBE (ID: 503)	03-Jan-2022	5/5 (100%)
Informed Consent - SBE (ID: 504)	04-Jan-2022	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	04-Jan-2022	5/5 (100%)
University of Arkansas at Little Rock (ID: 596)	04-Jan-2022	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

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COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)

COMPLETION REPORT - PART 2 OF 2
COURSEWORK TRANSCRIPT**

** NOTE: Scores on this Transcript Report reflect the most current quiz completions, including quizzes on optional (supplemental) elements of the course. See list below for details. See separate Requirements Report for the reported scores at the time all requirements for the course were met.

- **Name:** Kartik Sakthivel (ID: 10645833)
- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
- **Institution Email:** ksakthivel@ualr.edu
- **Institution Unit:** Computer and Information Sciences - Information Quality
- **Phone:** 6034939567

- **Curriculum Group:** Human Research
- **Course Learner Group:** Group 1.Social and Behavioral Research Investigators and Key Personnel
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 45783192
- **Report Date:** 05-Mar-2022
- **Current Score**:** 100

REQUIRED, ELECTIVE, AND SUPPLEMENTAL MODULES	MOST RECENT	SCORE
Defining Research with Human Subjects - SBE (ID: 491)	03-Jan-2022	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	03-Jan-2022	5/5 (100%)
Belmont Report and Its Principles (ID: 1127)	20-Dec-2021	3/3 (100%)
Assessing Risk - SBE (ID: 503)	03-Jan-2022	5/5 (100%)
Informed Consent - SBE (ID: 504)	04-Jan-2022	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	04-Jan-2022	5/5 (100%)
University of Arkansas at Little Rock (ID: 596)	04-Jan-2022	No Quiz
History and Ethical Principles - SBE (ID: 490)	20-Dec-2021	5/5 (100%)

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COMPLETION REPORT - PART 1 OF 2
COURSEWORK REQUIREMENTS*

* NOTE: Scores on this Requirements Report reflect quiz completions at the time all requirements for the course were met. See list below for details. See separate Transcript Report for more recent quiz scores, including those on optional (supplemental) course elements.

- **Name:** Elizabeth Pierce (ID: 501570)
- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
- **Institution Email:** expierce@ualr.edu
- **Institution Unit:** Information Science
- **Phone:** 501-916-5223

- **Curriculum Group:** Human Research
- **Course Learner Group:** Group 1.Social and Behavioral Research Investigators and Key Personnel
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 43846411
- **Completion Date:** 05-Sep-2021
- **Expiration Date:** 04-Sep-2024
- **Minimum Passing:** 100
- **Reported Score*:** 100

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
Belmont Report and Its Principles (ID: 1127)	03-Sep-2021	3/3 (100%)
History and Ethical Principles - SBE (ID: 490)	03-Sep-2021	5/5 (100%)
Defining Research with Human Subjects - SBE (ID: 491)	03-Sep-2021	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	04-Sep-2021	5/5 (100%)
Assessing Risk - SBE (ID: 503)	04-Sep-2021	5/5 (100%)
Informed Consent - SBE (ID: 504)	04-Sep-2021	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	05-Sep-2021	5/5 (100%)
University of Arkansas at Little Rock (ID: 596)	05-Sep-2021	No Quiz

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

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COMPLETION REPORT - PART 2 OF 2
COURSEWORK TRANSCRIPT**

** NOTE: Scores on this Transcript Report reflect the most current quiz completions, including quizzes on optional (supplemental) elements of the course. See list below for details. See separate Requirements Report for the reported scores at the time all requirements for the course were met.

- **Name:** Elizabeth Pierce (ID: 501570)
- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
- **Institution Email:** expierce@ualr.edu
- **Institution Unit:** Information Science
- **Phone:** 501-916-5223

- **Curriculum Group:** Human Research
- **Course Learner Group:** Group 1.Social and Behavioral Research Investigators and Key Personnel
- **Stage:** Stage 1 - Basic Course

- **Record ID:** 43846411
- **Report Date:** 26-Jan-2022
- **Current Score**:** 100

REQUIRED, ELECTIVE, AND SUPPLEMENTAL MODULES	MOST RECENT	SCORE
Defining Research with Human Subjects - SBE (ID: 491)	03-Sep-2021	5/5 (100%)
The Federal Regulations - SBE (ID: 502)	04-Sep-2021	5/5 (100%)
Belmont Report and Its Principles (ID: 1127)	03-Sep-2021	3/3 (100%)
Assessing Risk - SBE (ID: 503)	04-Sep-2021	5/5 (100%)
Informed Consent - SBE (ID: 504)	04-Sep-2021	5/5 (100%)
Privacy and Confidentiality - SBE (ID: 505)	05-Sep-2021	5/5 (100%)
University of Arkansas at Little Rock (ID: 596)	05-Sep-2021	No Quiz
History and Ethical Principles - SBE (ID: 490)	03-Sep-2021	5/5 (100%)

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COMPLETION REPORT - PART 1 OF 2
COURSEWORK REQUIREMENTS*

* NOTE: Scores on this Requirements Report reflect quiz completions at the time all requirements for the course were met. See list below for details. See separate Transcript Report for more recent quiz scores, including those on optional (supplemental) course elements.

- **Name:** John Talburt (ID: 1655553)
- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
- **Institution Email:** jrtalburt@ualr.edu
- **Institution Unit:** Information Science
- **Phone:** 501-916-5228

- **Curriculum Group:** Human Research
- **Course Learner Group:** Group 1.Social and Behavioral Research Investigators and Key Personnel
- **Stage:** Stage 4 - Refresher Course

- **Record ID:** 46944776
- **Completion Date:** 02-Mar-2022
- **Expiration Date:** 01-Mar-2025
- **Minimum Passing:** 100
- **Reported Score*:** 100

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
SBE Refresher 1 - Instructions (ID: 943)	02-Mar-2022	No Quiz
SBE Refresher 1 – History and Ethical Principles (ID: 936)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Federal Regulations for Protecting Research Subjects (ID: 937)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Informed Consent (ID: 938)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Defining Research with Human Subjects (ID: 15029)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Privacy and Confidentiality (ID: 15035)	02-Mar-2022	4/4 (100%)
SBE Refresher 1 – Assessing Risk (ID: 15034)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Research with Prisoners (ID: 939)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Research with Children (ID: 15036)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Research in Educational Settings (ID: 940)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – International Research (ID: 15028)	02-Mar-2022	2/2 (100%)

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COMPLETION REPORT - PART 2 OF 2
COURSEWORK TRANSCRIPT**

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- **Institution Affiliation:** University of Arkansas at Little Rock (ID: 401)
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- **Curriculum Group:** Human Research
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- **Stage:** Stage 4 - Refresher Course

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SBE Refresher 1 – Informed Consent (ID: 938)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Assessing Risk (ID: 15034)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Privacy and Confidentiality (ID: 15035)	02-Mar-2022	4/4 (100%)
SBE Refresher 1 – Research with Prisoners (ID: 939)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Research with Children (ID: 15036)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – Research in Educational Settings (ID: 940)	02-Mar-2022	2/2 (100%)
SBE Refresher 1 – International Research (ID: 15028)	02-Mar-2022	2/2 (100%)

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